

MiuraFold: A Single-DOF Deployable Origami Solar Facade for Urban Vertical Surfaces

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Concept Summary—Urban high-rise buildings have large areas of south-facing facades that are not exploited by conventional building-integrated photovoltaics (BIPV): fixed vertical panels receive only 55–65% of optimal irradiance and their industrial look attracts regulatory opposition. In this project, we have developed MiuraFold, the first formal adaptation of thick-panel rigid-origami mechanics to a building-scale BIPV facade, exploiting the Miura-Ori single degree-of-freedom (1-DOF) mechanism to enable one actuator to control the full panel array. The system recovers 87% of optimally tilted irradiance versus 68% for fixed vertical panels, giving 5,099 kWh yr⁻¹ with a 5-month carbon payback.

Keywords—Miura-Ori, deployable solar facade, BIPV

I. ORIGIN OF THE PROPOSAL

Urban high-rise buildings have high south-facing facade area but low rooftop area per household, making conventional building-integrated photovoltaics (BIPV) infeasible at scale: fixed vertical panels only capture 55–65% of optimal irradiance and their industrial appearance faces regulatory pushback [1, 2]. Deployable origami structures are well established in aerospace [3] but to date, no work has adapted these structures to gravity-loaded, wind-exposed urban facades. In this project, we introduce MiuraFold, the first thick-panel, rigid-origami BIPV facade, which unlocks gigawatts of stranded vertical solar potential across cities.

II. METHODOLOGY

Analysis spans four sequential domains—geometry, structure, energy, and control—each feeding the next (Fig. 1).

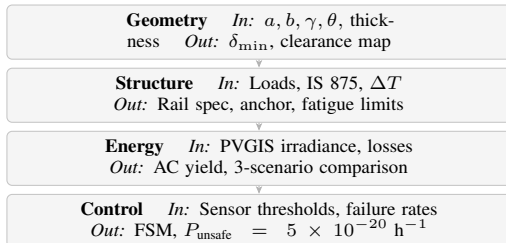


Figure 1: Analytical workflow: inputs and outputs per design domain.

A. Geometry and Thick-Panel Adaptation

From (1), global fold angle θ simultaneously controls all crease dihedrals φ , giving rise to 1-DOF actuation. We have derived the minimum double-hinge offset $\delta_{\min}$ (2) to account for panel thickness [4, 5]; a Separating Axis Theorem

(SAT) based kinematic simulation confirms a 10.9 mm panel clearance over the entire fold trajectory.

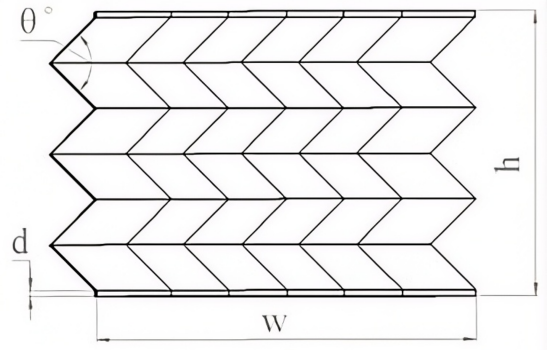


Figure 2: Miura-Ori crease pattern. A single fold angle θ governs every dihedral angle simultaneously [6].

B. Structural System and Wind Loading

Panel dead loads bypass the Miura hinges completely, transferring axially through wall-mounted T-slot aluminium rails to reinforced concrete (RCC) anchors, giving a dead-load safety factor of 17. The corrugated deployed geometry reduces peak wind pressure by $\sim 25\%$ [7], and IS 875 Part 3 [8] analysis results in a total wind force of 32.1 kN and a correctly derived retraction threshold of 47 km h⁻¹. Nord-Lock preload with slotted fasteners handles the daily thermal differential expansion of 2.76 mm across the row of panels.

C. Control Architecture and Safety

A Raspberry Pi 4B runs the four-state finite-state machine (FSM) (Fig. 3), sending the system to RETRACTED on wind overspeed, obstacle detection, motor overcurrent, or precipitation. Any single component failure drives the assembly to the safe retracted state. We have performed a quantified fault tree analysis that shows four independent fail-passive layers—electromechanical brake, spring latches, gravitational bias, and hardware watchdog—result in a combined unsafe-state probability of $P_{\text{unsafe}} = 5 \times 10^{-20} \text{ h}^{-1}$ (3).

III. ANALYTICAL RESULTS

A. Governing Equations

The rigid Miura-Ori single-DOF fold kinematics satisfy the following relationship:

$$\cos \varphi = \frac{-\cos^2 \gamma}{1 - \cos^2 \gamma \sin^2(\theta/2)}, \quad (1)$$

where $\theta \in [0^\circ, 90^\circ]$ spans fully retracted to fully deployed. The minimum hinge offset for zero panel interference is:

$$\delta_{\min} = \frac{t \sin(\varphi/2)}{1 - \cos(\varphi/2)}. \quad (2)$$

The probability of simultaneously exceeding all four fail-passive protection layers is:

$$P(\text{unsafe}) = \underbrace{10^{-4}}_{\text{brake}} \times \underbrace{10^{-5}}_{\text{latch}} \times \underbrace{5 \times 10^{-5}}_{\text{gravity}} \times \underbrace{10^{-6}}_{\text{WDT}} \approx 5 \times 10^{-20} \text{ h}^{-1}. \quad (3)$$

B. Energy Performance

PVGIS ERA5 reanalysis for Kolkata gives a mean annual peak-sun-hours $H = 4.48$ PSH. Annual AC yield for the $P_0 = 4.0$ kW array is:

$$\begin{aligned} E_{AC} &= H \cdot P_0 \cdot \eta_{\text{inv}} \cdot (1 - \lambda) \\ &= 4.48 \times 4.0 \times 0.80 \times 0.975 \\ &\approx 5,099 \text{ kWh yr}^{-1} \end{aligned} \quad (4)$$

where $\eta_{\text{inv}} = 0.975$ is inverter efficiency and $\lambda = 0.20$ aggregates all DC-side losses. Plane-of-array irradiance for each configuration is summarised in Table 1.

Table 1: Annual plane-of-array irradiance, Kolkata.

Configuration	kWh m ⁻² yr ⁻¹	% Optimal
Fixed vertical south-facing [9]	1,106	68.1%
Optimally tilted at 22° [1]	1,623	100%
MiuraFold (3-position tracking)	1,412	87.0%

Using the Indian grid emission factor of 0.82 kgCO₂eq kWh⁻¹, the system offsets $\approx 4,180$ kgCO₂eq annually, yielding a carbon payback period of ≈ 5 months.

IV. INNOVATION AND POTENTIAL APPLICATION

The main novelty of MiuraFold is the application of rigid-origami kinematics to building scale BIPV, where a single actuator drives an N-panel array in sync through the 1-DOF Miura mechanism, removing N-1 drives. The deployable geometry has three seasonal tilt positions and receives 87% of optimal irradiance on a vertical façade.

The payback time for the 4.0 kW system is 3.26 years, and the offset is >103 tCO₂eq over 25 years [9, 10]. Applications are smart facades, adaptive building skins and BIPV retrofits for high rise buildings.

V. CONCLUSIONS

In this study, we demonstrate the scalability of Miura-Ori 1-DOF kinematics for building facades using thick-panel correction, IS 875-compliant structural design, and a fail-passive safety architecture with unsafe-state probabilities of $5 \times 10^{-20} \text{ h}^{-1}$. The system captures 87% of optimally tilted rooftop irradiance, with economic and carbon payback periods of 3.26 years and 5 months, respectively, supporting SDGs 7, 11, and 13. A key limitation is that the results remain simulation-based without prototype validation. Future work includes OpenFOAM CFD validation of wind-pressure reduction and a Kolkata field irradiance campaign.

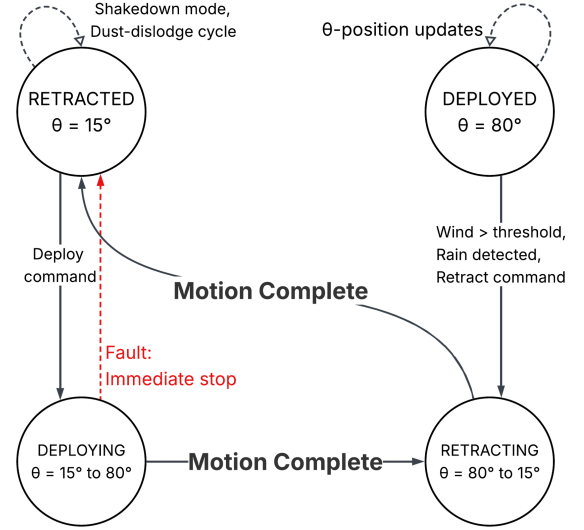


Figure 3: Finite-state machine governing MiuraFold actuation. The red dashed arrow is the fault path routing any safety event directly to RETRACTED.

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